

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)**  
**ACADEX PRACTICE EXAMINATION**  
**O-LEVEL · MATHEMATICS**

**MATHEMATICS 4004/1**

**PAPER 1 November 2018**

**2 hours 30 minutes**

**PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED**

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2018 4004/1.

**Additional materials:** Electronic calculators must NOT be used. Geometrical instruments may be used.

**INSTRUCTIONS:** Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Express 30 as a percentage of 60. [2]
  
2. Work out  $4/3 \div 2/3$ . [3]
  
3. Write 70000 in standard form. [2]
  
4. Overnight in Bulawayo the temperature was  $2^{\circ}\text{C}$ . By 14:00 it was  $24^{\circ}\text{C}$ . Find the rise. [2]
  
5.  $n(A) = 13$ ,  $n(B) = 11$ ,  $n(A \cap B) = 3$ . Find  $n(A \cup B)$ . [3]
  
6. Simplify  $(3^2)^2$ . Leave as a power of 3. [3]
  
7. Share \$22 in the ratio 4 : 7. [4]
  
8. Expand  $(x + 4)(x + 2)$ . [4]
  
9. Express 96 as a product of its prime factors. [4]

- 10.** Find the HCF and LCM of 12 and 42. **[4]**
- 11.** Make  $r$  the subject of  $C = 2\pi r$ . **[4]**
- 12.** Find the mean of 6, 12, 9, 12, 11. **[3]**
- 13.** Solve  $3x + 3 < 12$ . **[3]**
- 14.** Find the midpoint of (0, 8) and (3, 9). **[3]**
- 15.** Solve  $x + y = 6$  and  $x - y = 0$ . **[4]**
- 16.** The  $n$ th term is  $8 + 5(n - 1)$ . Find the 8th term. **[3]**
- 17.** A triangular plot has base 10 m and perpendicular height 5 m. Find its area. **[3]**
- 18.** Factorise  $x^2 - 16$ . **[4]**
- 19.**  $\tan 45^\circ = 1$ . In a right-angled triangle the adjacent side to  $45^\circ$  is 8 cm. Find the opposite side. **[4]**
- 20.** A cuboid water tank measures 12 m by 6 m by 5 m. Find its volume. **[3]**
- 21.** Solve  $6x + 7 = 25$ . **[3]**

- 22.** A ladder of length 15 m leans against a wall. The foot is 9 m from the wall. How far up the wall does it reach? **[4]**
- 23.** A shirt in Bulawayo is marked \$120. In a sale it is 25% off. Sale price? **[4]**
- 24.** Write the equation of the line with gradient 3 passing through (0, 4). **[3]**
- 25.** Factorise  $x^2 + 8x + 12$ . **[4]**
- 26.** In triangle ABC,  $\hat{A} = 55^\circ$  and  $\hat{B} = 69^\circ$ . Find  $\hat{C}$ . **[3]**
- 27.**  $u = (3 ; 0)$ ,  $v = (-2 ; 5)$ . Find  $2u + v$ . **[4]**
- 28.** A regular polygon has 6 sides. Find each exterior angle. **[3]**
- 29.** A bag contains 5 red, 5 blue and 5 green beads.  $P(\text{not green})$ ? **[3]**
- 30.** Circumference of a circle radius 21 cm. Take  $\pi = 22/7$ . **[4]**

**END OF QUESTION PAPER**

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).